

Goal

Remote monitoring of post-stroke patients for rehabilitation

- Wearable device records:
 - muscle activity, segment biomechanics (accel, forces)
 - **Detect:** fatigue, muscle growth, atrophy
- Cloud based system for remotely transmitting patient data to the clinician.

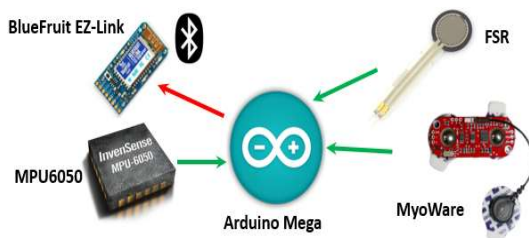
Background

- Around 700 000 strokes are reported annually in US [1]
- Stroke is the third leading cause of death, major loss of disability and lost productivity. [2]
- Lack of resources and availability of therapists for such a large number of patients results in early discharge from hospitals. [2], [3]
- Patients often regress after discharge
- Muscle fatigue affects stroke patients and their rehabilitation strategies drastically.

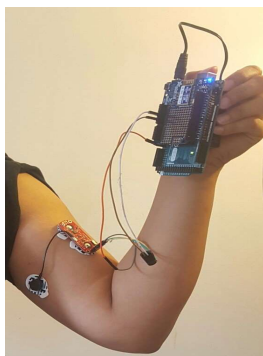
Introduction

- Fatigue is a feeling of exhaustion during physical or mental activity and is not ameliorated by rest [4]
- Impact: Fatigue affects rehabilitation by creating hindrance in regaining function lost after stroke [5]
- Solution
Physical exercise strengthens muscle and thus reduces muscle fatigue.

Device



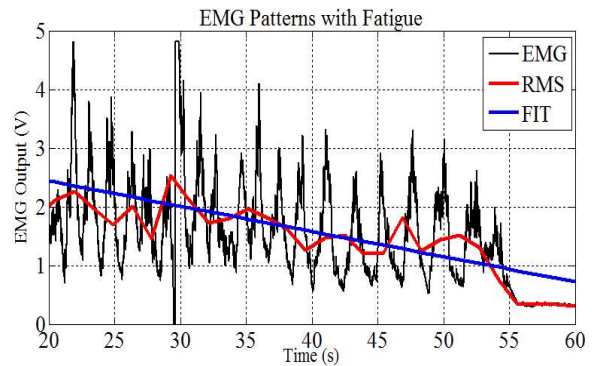
- Arduino based architecture (Arduino Mega)
- InvenSense Inc. MPU6050 (3 axis MEMs accelerometer and gyroscope) for motion tracking and gait analysis
- Myoware muscle sensor (Advancer Technologies LLC) for recording emg signal for muscle fatigue and growth detection.
- Force sensitive resistor placed on shoe-sole used for gait analysis.
- BlueFruit EZ-shield Bluetooth module for wireless data transmission



Methods

- A proof of concept experiments carried out on 24 year old male and 24 year old female.
- Repeated bicep curling with 10lb dumbbell and quadriceps extension using 30lb until the subject experienced weakness.
- Raw data filtered and rectified (Muscle Sensor V2)
- RMS value of EMG determined for trend (window size 1.5 seconds)
- Linear fit applied

Results



- EMG Signal decreases by 61%
- Figure shows the representative data for the female test subject obtained during the experiments.

Conclusion

- Classification of fatigue in time domain
- Provides a mechanism for wearables in at-home setting
- Fatigue correlations to strength and atrophy

Discussion

Part of a larger study to create cloud based monitoring platform for post-stroke patient rehabilitation

Applications in sports

- Determine optimized training
- Track muscle growth
- Quantifiable training parameters for athletes

References

- [1] N. F. Gordon, M. Gulanick, F. Costa, G. Fletcher, B. A. Franklin, E. J. Roth, and T. Shephard, "Physical Activity and Exercise Recommendations for Stroke Survivors," *Circulation*, vol. 109, no. 16, pp. 2031–2041, Apr. 2004.
- [2] S. Ruland, P. B. Gorelick, M. Schneck, D. Kim, C. G. Moore, and S. Leurgans, "Acute Stroke Care in Illinois A Statewide Assessment of Diagnostic and Treatment Capabilities," pp. 1334–1341, 2002.
- [3] M. Iosa, G. Morone, A. Fusco, M. Bragoni, P. Coiro, M. Multari, V. Venturiero, D. De Angelis, L. Pratesi, and S. Paolucci, "Seven capital devices for the future of stroke rehabilitation," *Stroke Res. Treat.*, vol. 2012, 2012.
- [4] M. Acciarresi, J. Bogouslavsky, and M. Paciaroni, "Post-stroke fatigue: epidemiology, clinical characteristics and treatment.," *Eur. Neurol.*, vol. 72, no. 5–6, pp. 255–261, 2014.
- [5] V. L. Barbour and G. E. Mead, "Fatigue after Stroke : The Patient 's Perspective," vol. 2012, 2012.